

Principles of Biology (11th Edition)
Department of Life Sciences

Chapter 4: Cellular Respiration & Energetics

Glycolysis Net Reaction: $\text{Glucose} + 2 \text{ NAD}^+ + 2 \text{ ADP} + 2 \text{ Pi} \rightarrow 2 \text{ Pyruvate} + 2 \text{ NADH} + 2 \text{ H}^+ + 2 \text{ ATP}$

Location: Cytosol

Net Yield: 2 ATP, 2 NADH per glucose molecule.

Chapter 9: The Citric Acid Cycle and Oxidative Phosphorylation

Substrate-Level vs Oxidative Phosphorylation

1. Pyruvate dehydrogenase complex converts pyruvate to Acetyl-CoA in the mitochondrial matrix.
2. Citric Acid Cycle produces 3 NADH, 1 FADH₂, and 1 GTP (ATP equivalent) per acetyl-CoA.
3. High-energy electron carriers feed into Electron Transport Chain Complexes I-IV.

Chapter 9, Section 4: Chemiosmotic Coupling & ATP Synthase

Peter Mitchell Chemiosmotic Hypothesis (Nobel Prize in Chemistry 1978)

Protons pumped across the inner mitochondrial membrane establish an electrochemical proton gradient ($\Delta\mu_{\text{H}^+}$)

The F₀ rotor subunit rotates as protons flow down the electrochemical gradient, mechanically driving ATP synthase

Total theoretical yield: 30 to 32 ATP molecules per oxidized glucose molecule.